



CS & IT ENGINEERING



THEORY OF COMPUTATION

Regular (DFA)

Lecture No. - 5



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Recap of Previous Lecture



Topic

Minimization of DFA

?????

① State equivalence method



Topics to be Covered



Topic

DFA

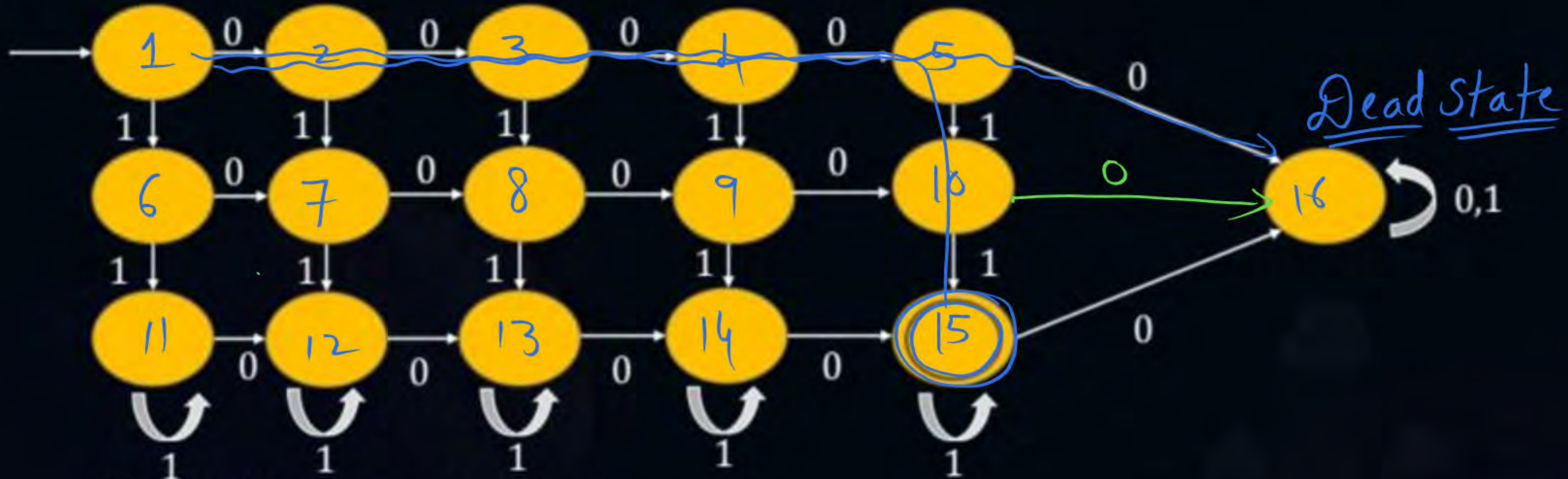
Construct

Topic

Product DFA

0's exactly 4 and # 1's atleast 2

#Q. Identify language accepted by following DFA



Set of all strings contains

00000000

A ☒ Length of the string atleast 6

B ☒ # 0's exactly 4 (and) 1's atleast 3

C ☒ # 0's atleast 4 and # 1's exactly 2

D ☒ None

MCQ



#Q. The number of binary strings of length n accepted by the following finite automata is-

Starting with 11

A $\{2^n | n \geq 0\}$

B $\{2^{n-1} | n \geq 1\}$

C $\{2^{n-2} | n \geq 2\}$

D $\{2^{n-3} | n \geq 3\}$

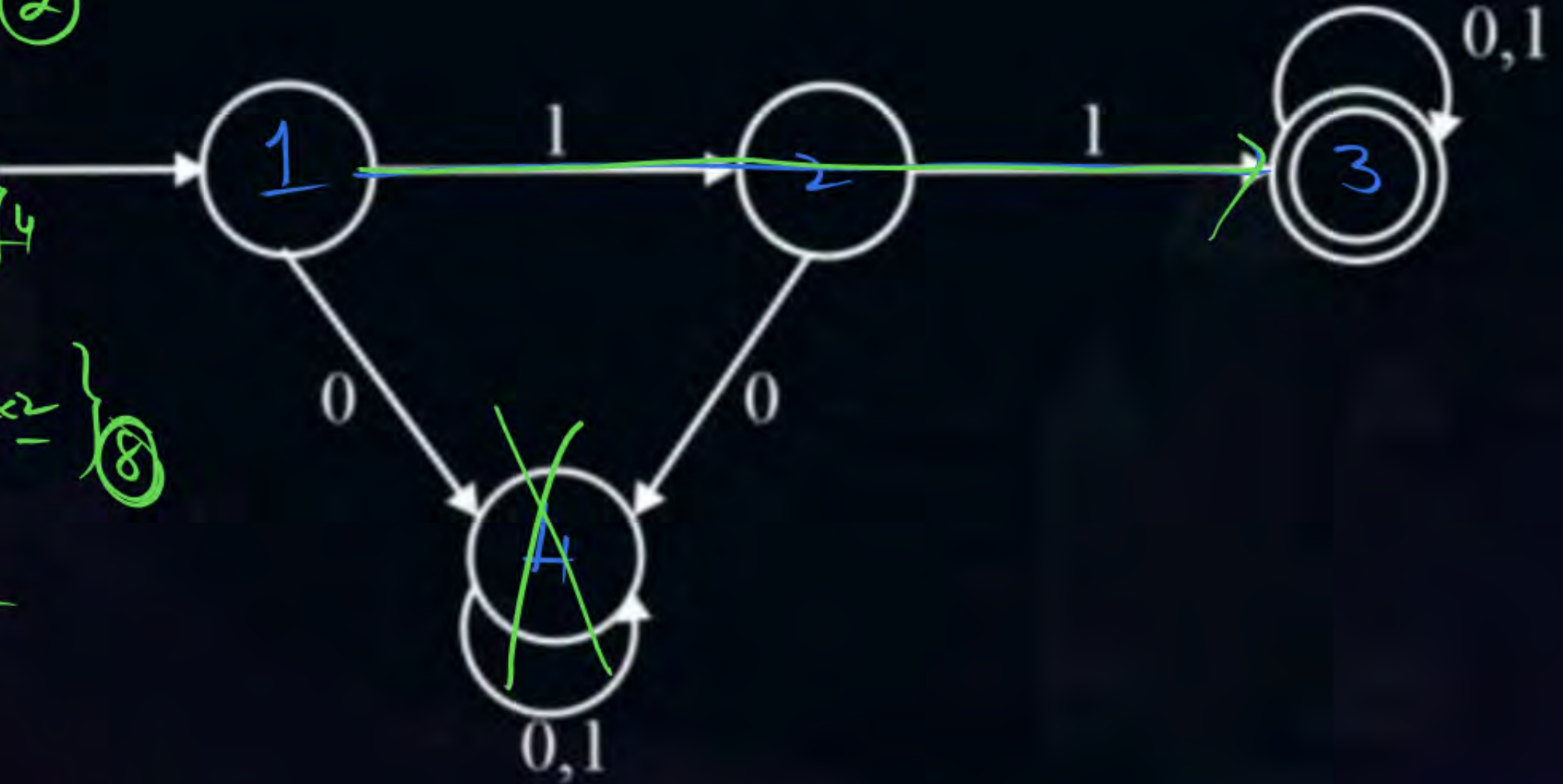
$n=2$
 $\{11\}$ ①

$n=3$
 $\{110, 111\}$ ②

$n=4$
 $\{1100, 1101, 1110, 1111\}$ ④

$n=5$
 $\{11000, 11001, 11010, 11011, 11100, 11101, 11110, 11111\}$ ⑧

$n=2$
 2

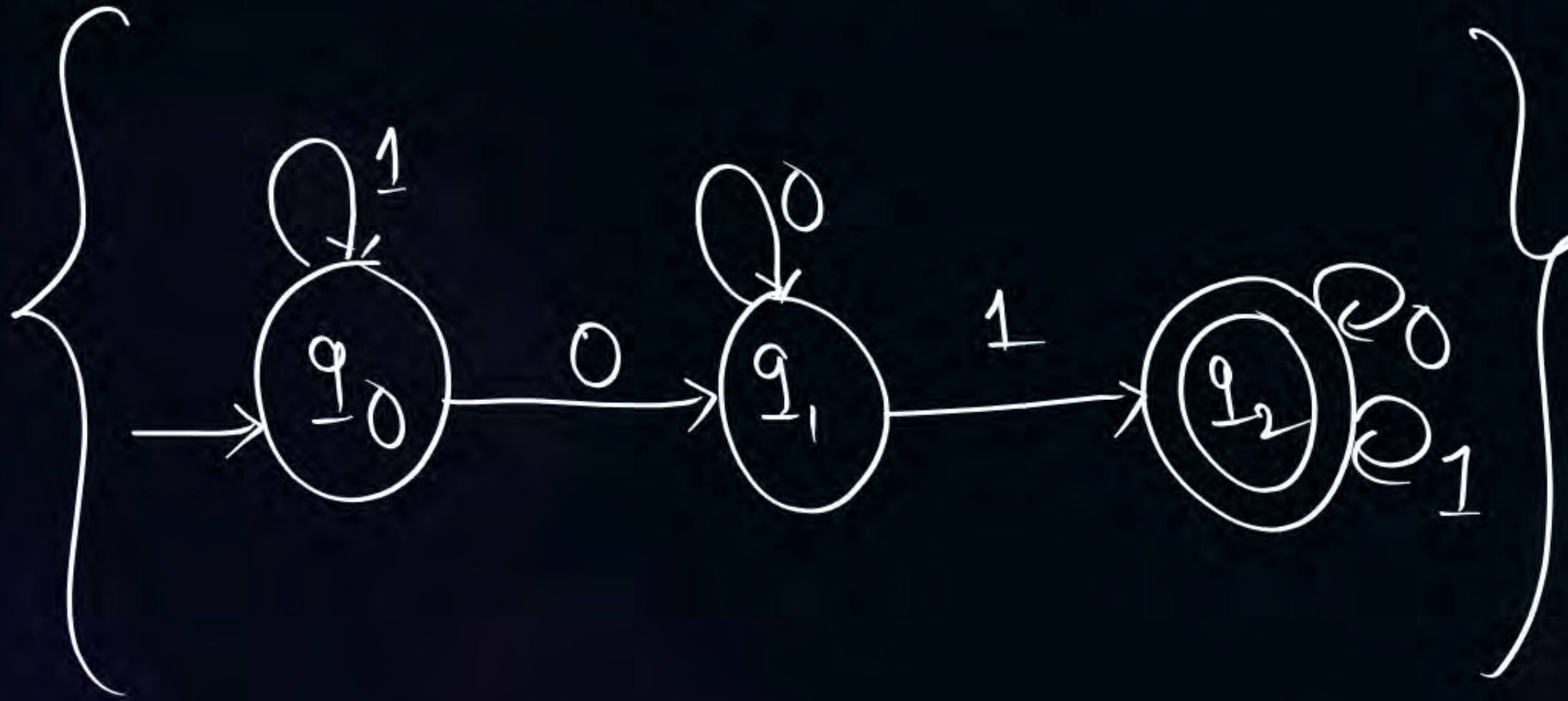


(Q) Construct minimal State DFA over $\Sigma = \{0, 1\}$ where each string having substring 01.

$L = \{01, 001, 101, 0011, \dots\}$

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \underline{01} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

0001
00001
111101

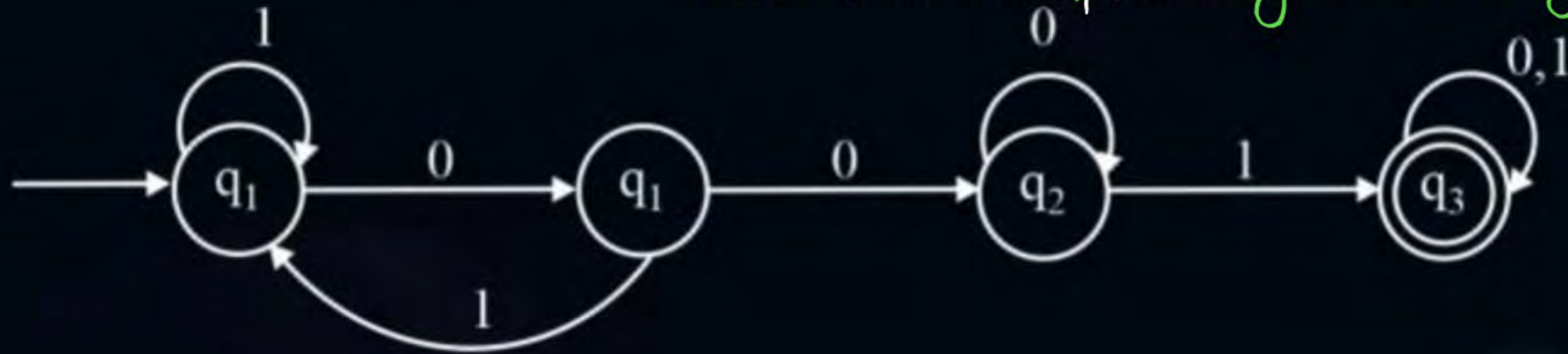


n length sub string
 $\Downarrow$
 $(n+1)$ states

MCQ



#Q. Consider the following deterministic finite automata m.



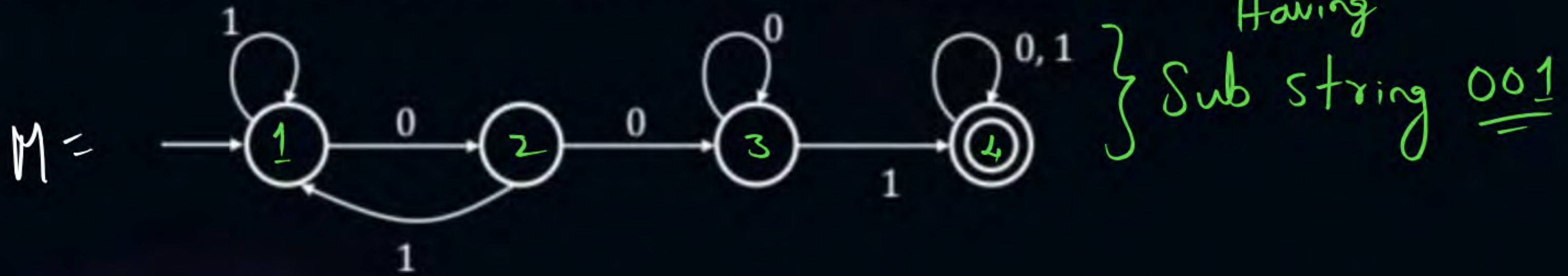
The language accepted by finite automata, which is obtained by interchanging final and non-final states in m, is-

- A** The set of all strings containing 001 as the substrings
- B** The set of all strings not containing 001 as the substrings
- C** The set of all strings ending with substrings 001
- D** The set of all strings ending with substrings 001

MCQ



#Q. Consider the following deterministic finite state automaton M.



16-7-

Let S denote the set of seven-bit binary strings in which the first, the fourth, and the last bits are 1. The number of strings in S that are accepted by M is

⑨ S: { 1 0 0 1 0 0 1 } 16 string of M

A 1

B 5

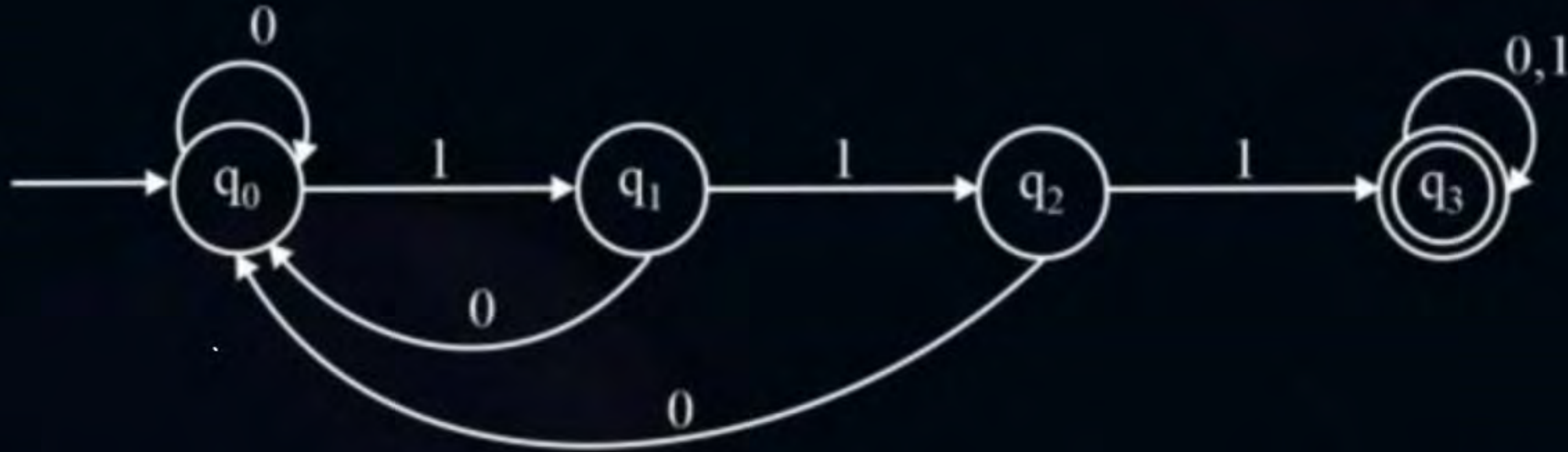
C 7 ✓

D 8

$$\begin{array}{l}
 5 \left\{ \begin{array}{l}
 | \quad \underline{00} \quad | \quad 00 \quad | \\
 | \quad \underline{00} \quad | \quad 01 \quad | \\
 | \quad \underline{00} \quad | \quad 10 \quad | \\
 | \quad \underline{00} \quad | \quad 11 \quad | \\
 | \quad 01 \quad | \quad \underline{00} \quad | \\
 | \quad 01 \quad | \quad 01 \quad | \\
 | \quad 01 \quad | \quad 10 \quad | \\
 | \quad 01 \quad | \quad 11 \quad |
 \end{array} \right.
 \end{array}$$

$$\begin{array}{l}
 | \quad 10 \quad | \quad \underline{00} \quad | \quad \} 6 \\
 | \quad 10 \quad | \quad 01 \quad | \\
 | \quad 10 \quad | \quad 10 \quad | \\
 | \quad 10 \quad | \quad 11 \quad | \\
 | \quad 11 \quad | \quad \underline{00} \quad | \quad \} 7 \\
 | \quad 11 \quad | \quad 01 \quad | \\
 | \quad 11 \quad | \quad 10 \quad | \\
 | \quad 11 \quad | \quad 11 \quad |
 \end{array}$$

#Q. Consider the finite automata m.



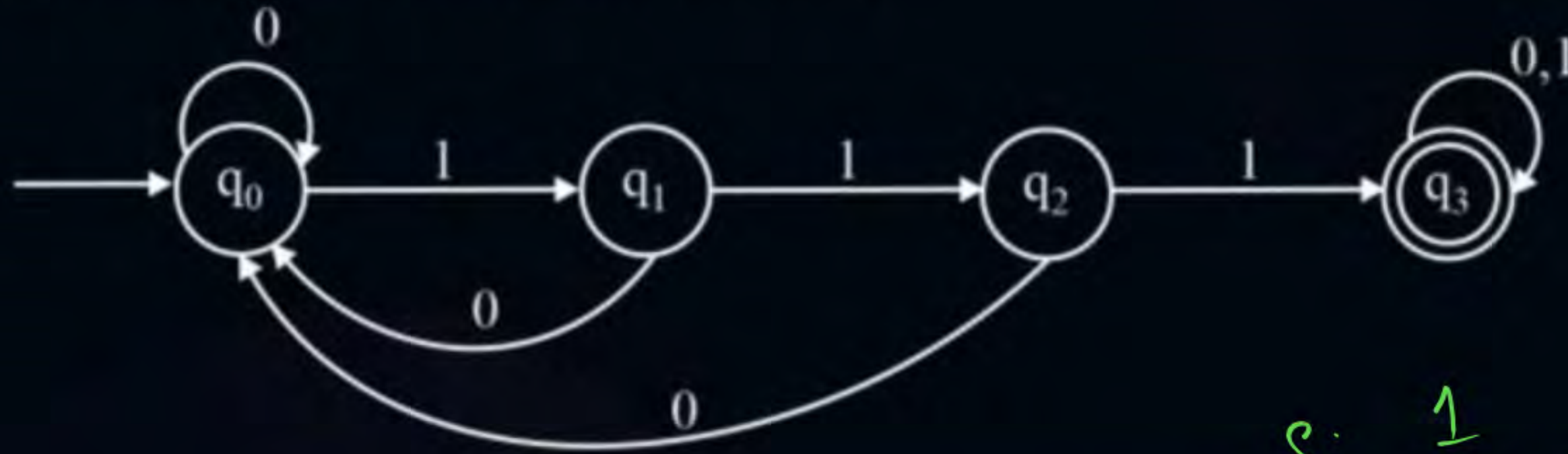
The language accepted by m is, over the alphabet $\{0,1\}$

- A** The set of all strings containing three consecutive I's
- B** The set of all strings not containing three consecutive I's
- C** The set of all strings beginning with three consecutive I's
- D** The set of all strings ending with three consecutive I's

MCQ



#Q. Consider the finite automata m.



Having
Sub string (111)

S: 1 _ _ 1 _ _ } 16

Let S denotes the set of all six bit binary strings in which first and fourth bits are 1. The number of strings in S that are accepted by m is-

A 1

B 4

C 7

D 8

	0	0		0	0	
	0	0		0	1	
	0	0		1	0	
	0	0		<u>1</u>	<u>1</u>	6
	0	1		0	0	
	0	1		0	1	
	0	1		<u>1</u>	<u>1</u>	0
	0	1		<u>1</u>	<u>1</u>	4

	1	0		0	0	
	1	0		0	1	
	1	0		1	0	
{		1	0		<u>1</u>	<u>1</u>
		<u>1</u>	<u>1</u>		0	0
		<u>1</u>	<u>1</u>		0	1
		<u>1</u>	<u>1</u>		1	0
		<u>1</u>	<u>1</u>		1	1

(Q) Construct minimal state DFA over $\Sigma = \{a, b\}$ where
length of each string is exactly 4.

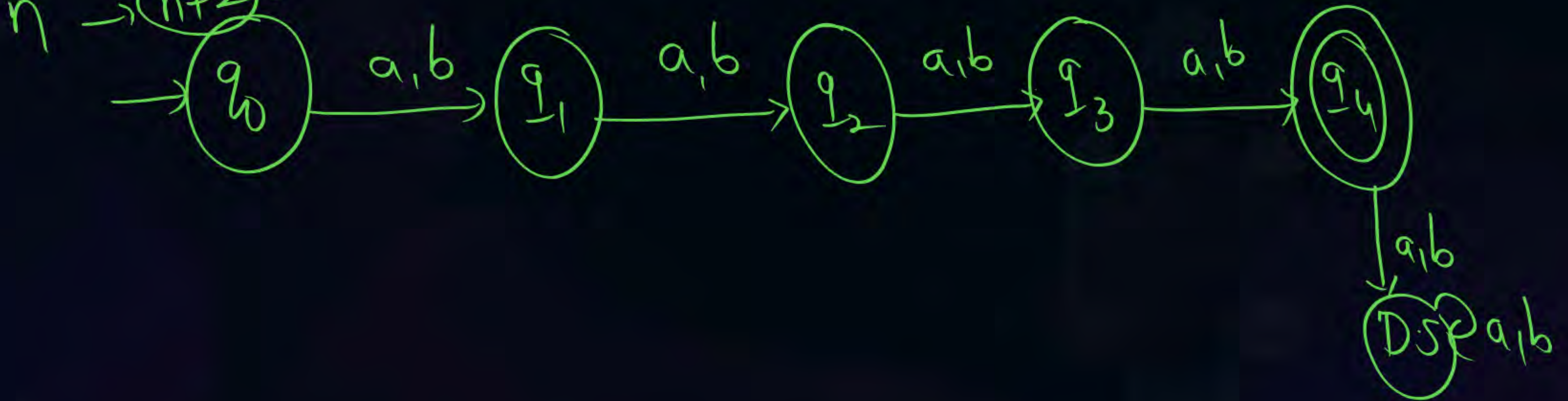
$\{aaaa, bbbb, abab, baba, \dots\}$

$\{\underline{2} \times \underline{2} \times \underline{2} \times \underline{2}\}$ 16 strings

6 States

exactly

4 → 6
5 → 7
n → (n+2) states



(1) Construct minimal state DFA over $\Sigma = \{a, b\}$ where length of the string is at most 4
 $\{0, 1, 2, 3, 4\}$

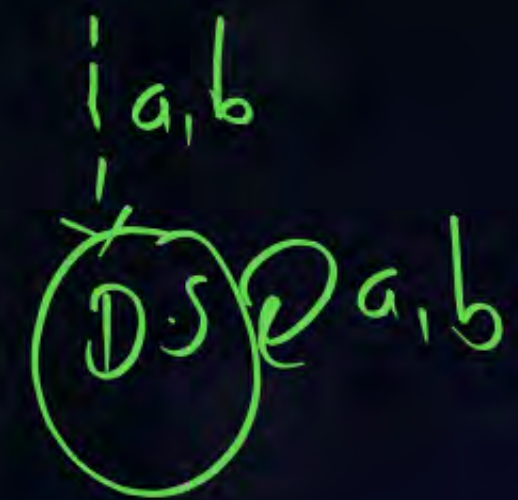


at most

4 $\rightarrow$ 6

5 $\rightarrow$ 7

$n \rightarrow (n+2)$ state



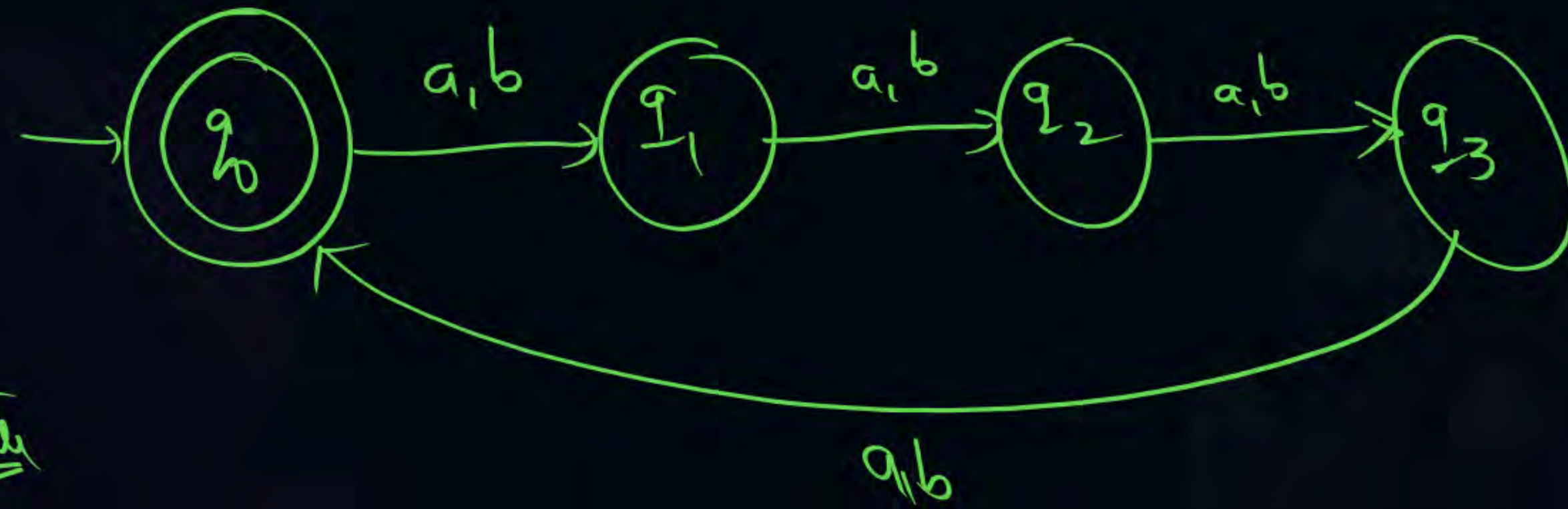
(Q) Construct minimal state DFA over $\Sigma = \{a, b\}$ where length of the string divisible by 4

$\{0, 4, 8, 12, \dots\}$
 $\underbrace{\quad}_4 \quad \underbrace{\quad}_4 \quad \underbrace{\quad}_4$

div by

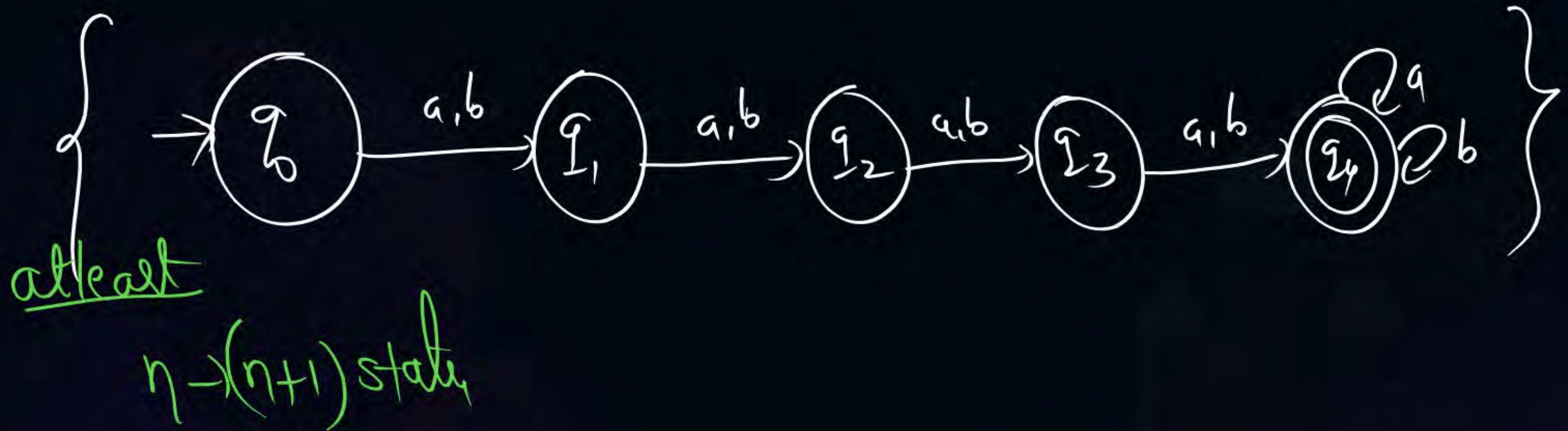
$4 \rightarrow 4$
 $5 \rightarrow 5$

$n \rightarrow n$ state



(Q) Construct minimal state DFA over $\Sigma = \{a, b\}$ where length of the string atleast 4.

5 state



(Q) Construct minimal state DFA over $\Sigma = \{a, b\}$ where

(Q) Construct minimal state DFA over $\Sigma = \{a, b\}$ where



2 mins Summary



Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU